

UNIT:-4 DIFFERENTIATIONS

APRIL

22

Friday

2022

Week-17
112-253

Q1 find the second order derivative of $e^{2x} \cdot \cos 3x + x^4$

Q2 Obtain the local maxima or local minima of $f(x) = x^3 - 6x^2 + 9x + 15$
Also find the local maximum or local minimum values $f(x)$

Q3 find the differential coefficient of $\sin^{-1}x$ from the first principle

Q4 If $x^m y^n = (x+y)^{m+n}$ then evaluate $\frac{dy}{dx}$

Q5 prove that $\text{sech}^2 x + \tanh^2 x = 1$

Q6 find $\frac{dy}{dx}$ when $y = x^{\sin x} + (\sin x)^x$

Q7 If $y = \sqrt{\cos x} + \sqrt{\cos x} + \sqrt{\cos x} + \dots \infty$

prove that

Q8 find $\frac{dy}{dx}$

a) $y = \frac{\sin x + x^2}{\cot 2x}$ b) $y = \tan^{-1}(\sqrt{1+x^2} + x)$

Never seem wisher or more learned than your company.

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May

APRIL

23

Saturday

2022

Week-17
113-235

Q9 If $y = (\sin^{-1} x)^2$ prove that $(1-x^2) \frac{d^2 y}{dx^2} = x \frac{dy}{dx} + 2$

Q10 find $\frac{dy}{dx}$ of any two of the following

Q a $y = \frac{x \tan x}{\sec x + \tan x}$

Q b $y = \sqrt{\sin \sqrt{\sin \sqrt{x}}}$

Q11 If $y = e^x (\sin x + \cos x)$ prove that $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2 = 0$

Q12 prove that $\sqrt{3} \sin x + 3 \cos x$ has a maximum value at $x = \pi/6$

Q13 find $\frac{dy}{dx}$ of any two of the following.

24 Sunday

Q a $y = \tan^{-1} \left(\frac{\sqrt{1+\cos x}}{\sqrt{1-\cos x}} \right)$

Q14 find the maximum and minimum value

$x^3 + 6x^2 + 9x + 15$

Q15 find $\frac{dy}{dx}$ where $y = (\sin x)^x + (\cos x)^x$

No greater shame to man than inhumanity.

April

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2022

Week-18
115-250

APRIL

Monday

25

Q16 find the maximum and minima of the function.

$$2x^3 - 15x^2 + 36x + 11$$

Q17 If $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$ Show that $\frac{dy}{dx} = -1$ at $\theta = \pi/2$

Q18 If $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ find $\frac{dy}{dx}$

Q19 If $\sin y = x \sin(a+y)$ then prove that

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

Q20 If $y = (\tan x)^{\cot x}$, find $\frac{dy}{dx}$

Q21 find the differential coefficient of the function $\sec^{-1}x$ from the first principle.

Q22 If $\sin y = \cos x$ find $\frac{dy}{dx}$ of $y = (x)^{\cos x} + (\sin x)^{\tan x}$

Q23 If $y = (\sin x)^{(\sin x)^{(\sin x) \dots \infty}}$ Show that

$$\frac{dy}{dx} = \frac{y^2 \cot x}{(1 - y \log_e \sin x)}$$

No man can serve two masters.

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

May

APRIL

26

Tuesday

2022

Week-18
115-249

Q24 find differential co-efficient of $\operatorname{cosec}^{-1} x$

Q25 find the maxima and minima of the function $2x^2 - 4x - 8$

Q26 If $y = \sqrt{\tan x} + \sqrt{\tan x} + \sqrt{\tan x} + \dots \infty$
prove that $\frac{dy}{dx} = \frac{\sec^2 x}{2y-1}$

Q27 If $y = e^{x \log_e(\sin 2x)}$ find $\frac{dy}{dx}$

Q28 If $y = (\tan^{-1} x)^2$ prove that
 $(1+x^2) y_2 + 2x(1+x^2) y_1 = 2$

Q29 If $u = e^{xyz}$ show that
 $\frac{d^3 u}{dx \cdot dy \cdot dz} = e^{xyz} [1 + 3xyz + x^2 y^2 z^2]$

Q30 find the maximum and minimum values of
 $x^3 - 2x^2 + x + 6$

No pains, no gains.

April

F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

* find $\frac{dy}{dx} = ?$

(i) $xy = x^3 + y^3$ Ans $3x^2 - y / x - 3y^2$

(ii) $x^m y^n = (x+y)^{m+n}$ Ans y/x

(iii) $x = y \log(xy)$ Ans $y(x-y) / x(x+y)$

(iv) $x^n + y^n = a^n$ Ans $-x^{n-1} / y^{n-1}$

(v) $x^{2/3} + y^{2/3} = a^{2/3}$ Ans $-y^{1/3} / x^{1/3}$

(vi) $y = (x+y)^2$ Ans $2(x+y) / 1-2x-2y$

(vii) $y = \sin(x+y)$ Ans $\cos(x+y) / 1 - \cos(x+y)$

(viii) $xy = \sin(x+y)$ Ans $\cos(x+y) - y / x - \cos(x+y)$

(ix) $\log y = \sin(x+y)$ Ans $y \cos(x+y) / 1 - y \cos(x+y)$

(x) $y = \sqrt{x + \sqrt{x + \sqrt{x} \dots \infty}}$ Ans $1/2y - 1$

(xi) $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x \dots \infty}}}$ Ans

(xii) $x^m y^n = (x-y)^{m+n}$ Ans y/x

(xiii) $x/y = \operatorname{cosec} xy$ Ans $\frac{\sin(xy) + xy \cos(xy)}{1 - x^2 \cos(xy)}$

(xiv) $x^2 y^2 = \sin(xy)$ Ans $-y/x$

(xv) $e^{xy} + xy = 0$ Ans $-y/x$

2022

APRIL

Week-18
117-248

Wednesday

27

Q31 find the differential coefficient of $\sec^{-1} \theta$ from the first principle